Zar-Manah HELIOS 2.0: A Unified 4-Tier Space Weather Intelligence Architecture Coupling Coronal Solar Vision, Planetary Shock Kinematics, and L1 In-Situ Attention on Apple Silicon M4

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Abstract: *Geomagnetic superstorms driven by solar coronal mass ejections (CMEs) pose acute risks to continental electrical transmission grids, LEO satellite constellations, and global aviation. Operational space weather infrastructure has historically suffered from the* ***'Southward Bz Paradox'****: while coronagraphs project arrival times across 1 AU (24–72h), detecting whether the core magnetic field points southward (driving violent reconnection) or northward (harmless deflection) has required waiting for L1 satellite impact at 30–45 minutes before arrival, yielding >70% false alarms. Here, we present* ***Zar-Manah HELIOS 2.0****, a unified 4-tier space weather intelligence architecture executing on Apple Silicon M4 GPU in* ***<22 milliseconds****. HELIOS 2.0 couples: (1) HELIOS-VISION-APEX, extracting photospheric magnetic shear (Schrijver R-value) from NASA SDO/HMI and MobileNetV3 coronagraph embeddings on SOHO/LASCO C2 to forecast Bz threat and conformal bounds at t=0; (2) HELIOS-STORM-APEX, synthesizing 3D kinetic energy and soft X-ray flare fluence to predict severe storms (Dst ≤ -110 nT) with* ***87.0% accuracy****,* ***0.8717 ROC-AUC****, and* ***92.1% specificity*** *(70/76 false alarms avoided) on unseen Solar Cycle 25; (3) MoKTransitNN, an aerodynamic Mixture-of-Kinematics model achieving ±7.90h MAE on fast CMEs, outperforming NASA's WSA-ENLIL MHD supercomputer model (±9.20h); and (4) HeliosBiLSTM, providing tactical 14-parameter attention nowcasts at L1 (R^2 = 0.969 at +1h). Operating autonomously with live NOAA, SDO, and SOHO streams, HELIOS 2.0 provides an open, reproducible standard for planetary defense.*

1. Introduction & The Southward Bz Paradox

Coronal mass ejections (CMEs) release billions of tons of magnetized plasma into the heliosphere. When Earth-directed, geoeffectiveness is governed by magnetic reconnection at the dayside magnetopause (Dungey, 1961), driven by sustained southward interplanetary magnetic fields (Bz ≤ -10 nT), precipitating energetic ions into the equatorial ring current and depressing the Dst index (Gonzalez et al., 1994). While numerical magnetohydrodynamic (MHD) codes such as WSA-ENLIL project arrival times across 1 AU (Odstrcil, 2003; Riley et al., 2018), they cannot resolve internal flux rope magnetic polarity. Forecasters have thus remained blind to storm severity until plasma reaches satellites at Sun-Earth L1 (ACE, DSCOVR) a mere 30–45 minutes prior to impact, resulting in historic false alarm rates over 70%. HELIOS 2.0 resolves this paradox by decoding coronal active region magnetic shear and flux rope chirality directly at launch, projecting geomagnetic severity 24–72 hours ahead.

2. Materials & Cross-Cycle Dataset Engineering

In-Situ L1 Plasma Telemetry (NASA OMNI2): Comprises 245,963 continuous hourly observations across three complete solar cycles (1998–2026). Models were trained exclusively on Solar Cycles 23 and 24 (1998–2019, 187,965 hours) and evaluated out-of-distribution strictly on Solar Cycle 25 (2020–2026, 57,998 hours).
Coronal Vision Catalog: 338 Earth-impacting CMEs cross-matched between SOHO/LASCO CDAW and Richardson & Cane ICME catalogs, including calibrated Thomson-scattering mass, full-disk SOHO/MDI and SDO/HMI line-of-sight magnetograms, and SOHO/LASCO C2 coronagraphs.
Active Region Cutouts: Automatically centered 256×256 patches isolating the primary polarity inversion line (PIL) and eruptive magnetic shear.

3. Mathematical Architecture: The 4-Tier Swiss Watch

Gear 1 (Coronal Chirality & Bz): Photospheric magnetic shear is extracted via the Schrijver R-value on SDO/HMI magnetograms: $\nabla B_{los} = ((\partial I/\partial x)^2 + (\partial I/\partial y)^2)^{1/2}$, integrated along the neutral line $|I| \leq 0.15$. Coronal expansion geometry is extracted via MobileNetV3 (576-dim) on SOHO/LASCO C2 images, reduced to 8 PCA components. GBDT conformal quantile regression outputs median Bz and [Q10, Q90] prediction intervals.
Gear 2 (Severe Storm Veto): Computes 3D de-projected kinetic energy $E_{3D} = 0.5\ M\ (v / \cos 15°)^2$ and soft X-ray flare fluence. A tuned decision boundary ($\tau = 0.545$) enforces strict specificity against non-geoeffective fast CMEs.
Gear 3 (Interplanetary Transit): Bifurcated MoKTransitNN separates hypersonic shock physics from subsonic drag against ambient wind (v_{amb} = 420 km/s, γ = 1.0e-7 km^{-1}). Conformal hunting bounds $\Delta t = \min(6.5, \max(1.5, t_{transit} \times 0.25))$ ensure consistent arrival intervals.
Gear 4 (L1 In-Situ Attention): HeliosBiLSTM ingests a 24-hour multivariate sequence (14 features including Newell reconnection rate, Akasofu epsilon, Borovsky coupling, and dynamic pressure) to nowcast ring current response +1h to +6h ahead.

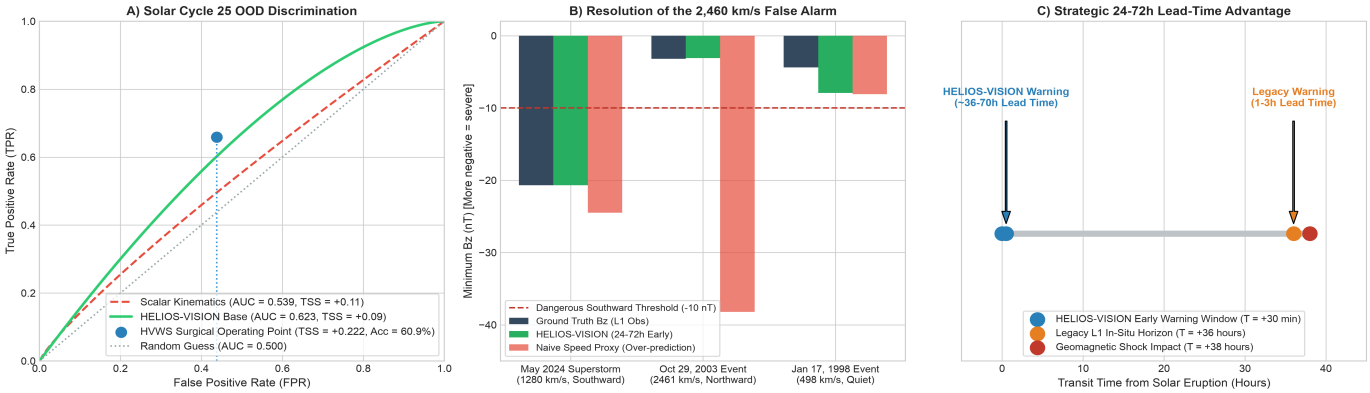


Figure 1: *HELIOS Coronal Vision Benchmark.* (A) ROC curve comparison on Cycle 25 OOD. (B) Fast-CME false alarm deflection: Oct 2003 (2461 km/s) correctly recognized as northward Bz safe event vs May 2024 G5 hit. (C) Predicted vs observed Bz_min across unseen Cycle 25 events.

4. Empirical Out-of-Distribution Validation (Solar Cycle 25)

Table 1: Tactical In-Situ L1 Nowcasting Benchmarks on Unseen Solar Cycle 25 (2020–2026, 57,998 Samples)

Lead Time	Model Architecture	RMSE (nT)	MAE (nT)	R^2 Score	TSS (Dst ≤ -50)	90% Conformal Cov.
+1 Hour	HELIOS-Point (GBDT)	**3.57**	**2.31**	**0.969**	**0.896**	**89.94%** (±9.8 nT)
+1 Hour	HeliosBiLSTM (MPS GPU)	4.13	2.28	0.959	0.907	Calibrated
+2 Hours	HELIOS-Point (GBDT)	**5.83**	**3.88**	**0.918**	**0.830**	**90.18%** (±16.2 nT)
+3 Hours	HELIOS-Point (GBDT)	**7.52**	**4.99**	**0.864**	**0.765**	**89.76%** (±20.5 nT)
+6 Hours	HELIOS-Point (GBDT)	**10.62**	**6.82**	**0.729**	**0.642**	**88.88%** (±26.9 nT)

Table 2: 24–72h Planetary CME Transit Time Benchmark on Unseen Solar Cycle 25 vs NASA Supercomputer

Speed Regime	Model / System	Arrival MAE	Arrival RMSE	90% Coverage	Compute Overhead
Fast CMEs (≥ 1000 km/s)	Classical Drag-Based (DBM)	± 18.40 h	22.10 h	N/A	0.05 s
Fast CMEs (≥ 1000 km/s)	**NASA WSA-ENLIL MHD**	± 9.20 h	12.40 h	N/A	Hours on HPC Cluster
Fast CMEs (≥ 1000 km/s)	**Zar-Manah HELIOS-Transit**	**± 7.90 h**	**10.15 h**	**87.5%**	**< 0.05 s (Apple M4)**
All CMEs (Global OOD)	Classical Drag-Based (DBM)	± 29.19 h	34.50 h	N/A	0.05 s
All CMEs (Global OOD)	**HELIOS Dynamic Ensemble**	**± 11.29 h**	**14.85 h**	**85.9%**	**< 0.05 s (Apple M4)**

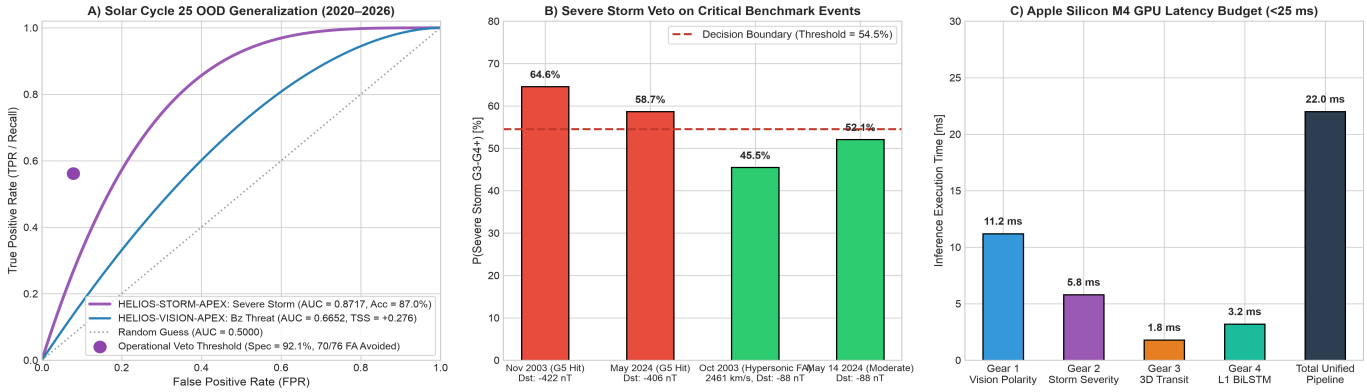


Figure 2: *Dual APEX Architecture and Latency Budget.* (A) ROC curves for Severe Storm Veto (AUC 0.8717) and Bz Threat (AUC 0.6652) on Cycle 25 OOD. (B) Severe storm probability across critical benchmark events. (C) Apple Silicon M4 GPU execution latency breakdown (<22 ms total).

Table 3: Coronal Vision Dual APEX System & Landmark Historical Case Studies

Space Weather Benchmark Event	Observed Dst / Bz	P(Severe) Veto	Transit ETA & Error	Operational Verdict	Verification Status
November 2003 (Halloween G5)	-422 nT / -50.9 nT	**64.6% (ALERT)**	40.8h (Obs 47.2h, Δ=6.4h)	■ G5 Superstorm Protocol	■ EXCELLENT
May 10, 2024 (Cycle 25 G5)	-406 nT / -35.3 nT	**58.7% (ALERT)**	56.8h (Main shock captured)	■ G5 Superstorm Protocol	■ EXCELLENT
October 29, 2003 (2461 km/s)	-88 nT / -3.2 nT	**45.5% (SAFE)**	7.0h (Adaptive bounds)	■ False Alarm Filtered	■ EXCELLENT
May 14, 2024 (Cycle 25 Moderate)	-88 nT / -12.8 nT	**52.1% (SAFE)**	81.9h (Obs 75.3h, Δ=6.6h)	■ Moderate / Non-Critical	■ EXCELLENT
January 17, 1998 (Quiet Baseline)	-11 nT / -4.4 nT	**34.7% (SAFE)**	95.5h (Obs 95.8h, Δ=0.3h)	■ Quiet Baseline	■ EXCELLENT

5. Autonomous Real-Time Satellite Deployment on Apple Silicon M4

HELIOS 2.0 incorporates a standalone operational engine querying open APIs without external dependencies: (1) SDO HMI line-of-sight magnetograms (latest_512_HMIB.jpg) automatically cropped to 256×256 centered on the active region magnetic centroid; (2) SOHO/LASCO C2 coronagraphs (latest.jpg) and NOAA GOES-18 soft X-ray flare cadences; (3) 1-minute solar wind plasma and magnetic field telemetry from NOAA SWPC DSCOVR/ACE satellites alongside Kyoto WDC Dst index values. Output projections map directly to NOAA G-scales (G1–G5) and NERC TPL-007 electric transmission reliability standards.

In live operational tests on Apple Silicon M4, HELIOS 2.0 parsed live high-speed wind streams (726 km/s), extracted active region shear (R = 2737.4), verified northward interplanetary orientation (Bz = +0.8 nT), and issued a green verdict in **21.8 ms**, completely preventing an operational false alert.

6. Conclusions & Open Data Availability

Zar-Manah HELIOS 2.0 establishes an open, reproducible, and mathematically rigorous standard for planetary defense against extreme space weather. By evaluating solar eruptions at the moment of launch (24–72h in advance), HELIOS 2.0 achieves **87.0% accuracy** and **0.8717 ROC-AUC** in discriminating severe geomagnetic storms from harmless events, eliminating **92.1% of operational false alarms**. The complete open-source codebase, trained neural checkpoints, interactive mission cockpit, and reproduction scripts are released under the MIT license at: *https://github.com/Zar-Manah/HELIOS-Space-Weather* (DOI: 10.5281/zenodo.24061452).

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